Holy Spirit High School **Name** ________________________________

AP Calculus AB

Summer Review Packet

Topic A: Functions

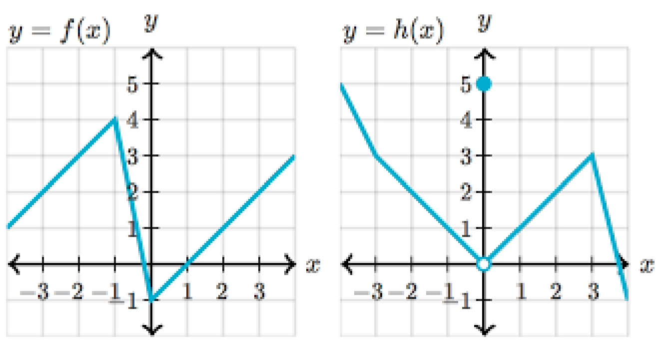
1.) If $f(x)=4x-x^2$, find:

a.) $f(4)-f(-4)$ b.) $\sqrt{f\left(\frac{3}{2}\right)}$ c.) $\dfrac{f(x+h)-f(x)}{2h}$

2.) If $V(r)=\frac{4}{3}\pi r^3$, find:

a.) $V\left(\frac{3}{4}\right)$ b.) V(r + 1) c.) $\dfrac{V(2r)}{V(r)}$

3.) If $f(x)$ and $g(x)$ are given in the graph, find:



a.) *(f – h) (3)* b.) *f(h(3))*

4.) If $f(x)=\begin{cases} -x, & x<0 \\ x^2-1, & 0\le x<2 \\ \sqrt{x+2}-2, & x\ge 2 \end{cases}$, find:

a.) $f(0)-f(2)$ b.) $\sqrt{5-f(-4)}$ c.) $f(f(3))$

Topic B: Domain and Range

Find the domain of the following functions using interval notation:

1.) $f(x) = 3$

2.) $y = x^3 - x^2 + x$

3.) $y = \dfrac{x^3 - x^2 + x}{x}$

4.) $y = \dfrac{x-4}{x^2 - 16}$

5.) $f(x) = \dfrac{1}{4x^2 - 4x - 3}$

6.) $y = \sqrt{2x-9}$

7.) $y = \log(x - 10)$

8.) $y = \dfrac{\sqrt{2x+14}}{x^2 - 49}$

Find the range of the following functions:

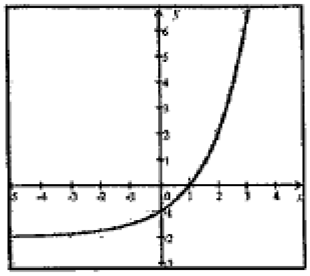
9.) $y = x^4 + x^2 - 1$

10.) $y = 100^x$

11.) $y = \sqrt{x^2+1} + 1$

Find the domain and range of the following functions using interval notation.

12.)



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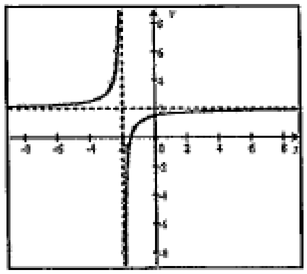

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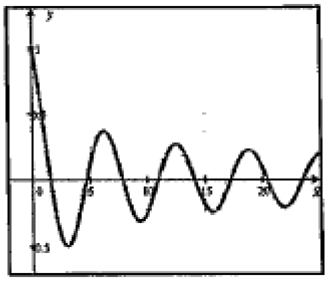
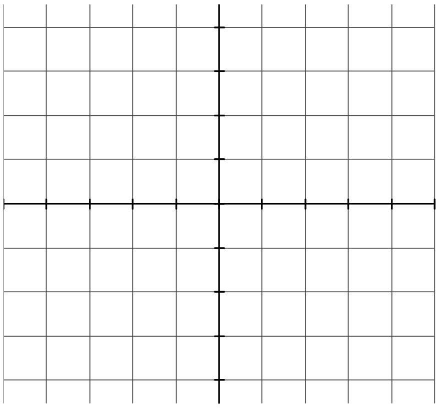


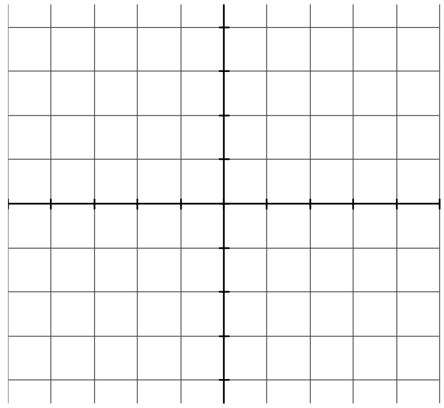
Topic C: Graphs of Common Functions

Sketch each of the following as accurately as possible. **You will need to be VERY familiar with each of these graphs throughout the year**. You may use a graphing calculator for some of them or you can use www.desmos.com. There is an app for Desmos as well that is free that you can install on your phones. Again, these are VERY important graphs to know. Be very accurate with regards to "open circles" and "closed circles" as those features may not be revealed on a graphing utility.

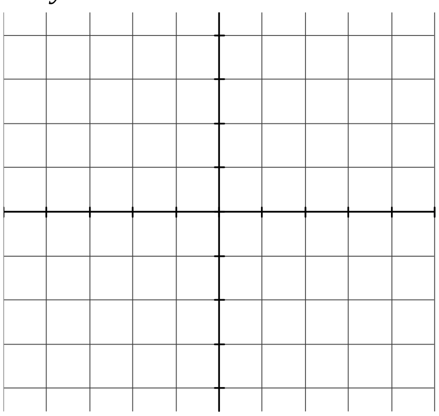
1. $y = x$



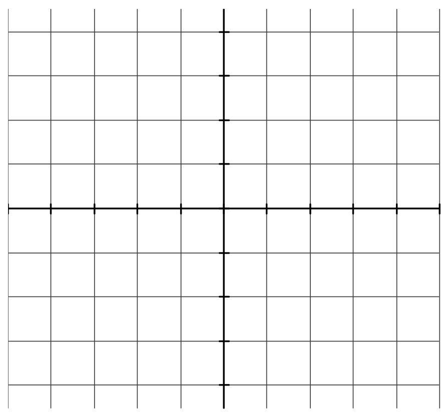
2. $y = x^2$



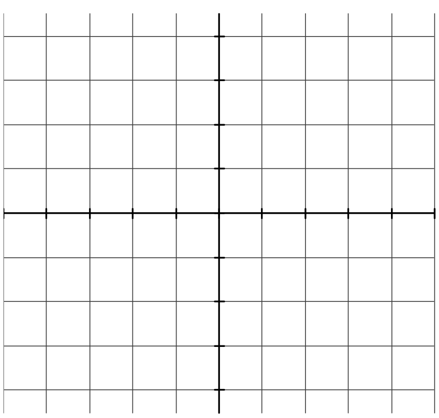
3. $y = x^3$



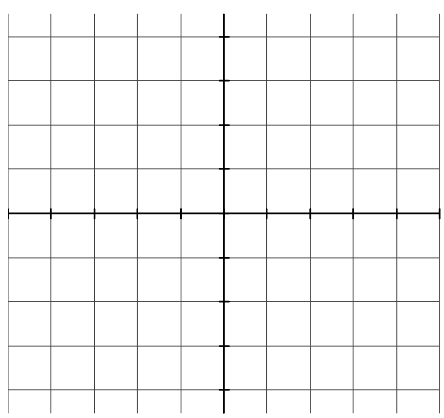
4. $y = \sqrt{x}$



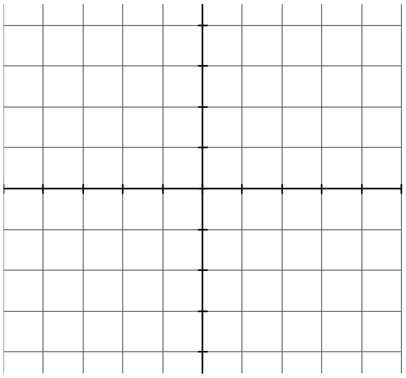
5. $y = |x|$



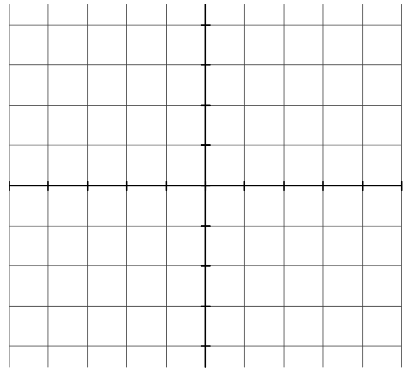
6. $y = \dfrac{|x|}{x}$



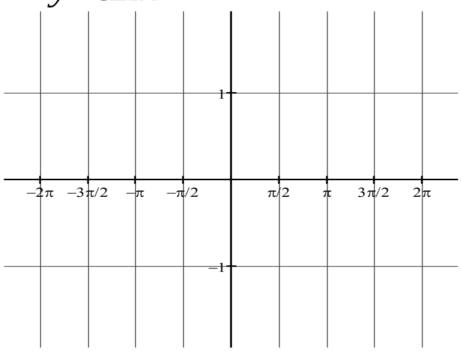
7. $y = x^{1/3}$



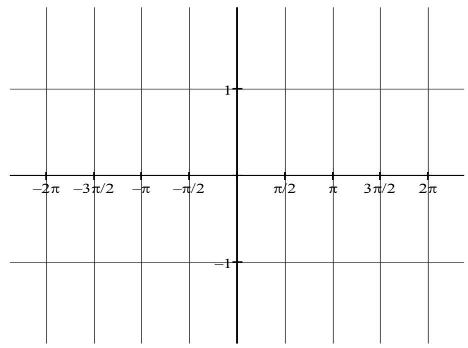
8. $y = x^{2/3}$



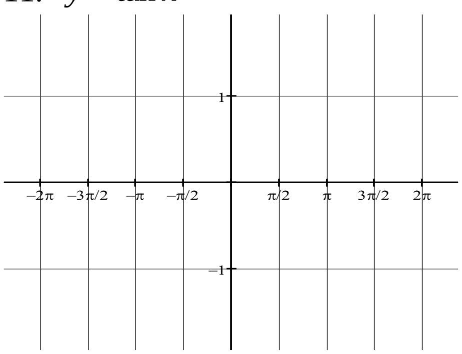
9. $y = \sin x$



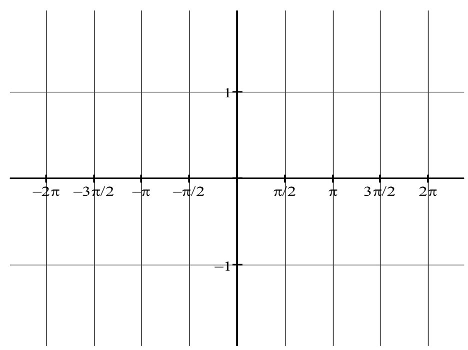
10. $y = \cos x$



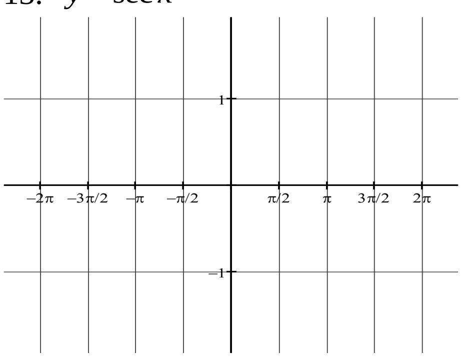
11. $y = \tan x$



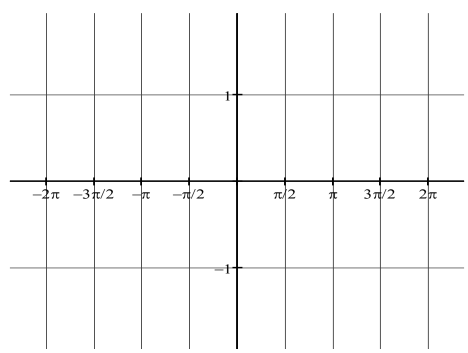
12. $y = \cot x$



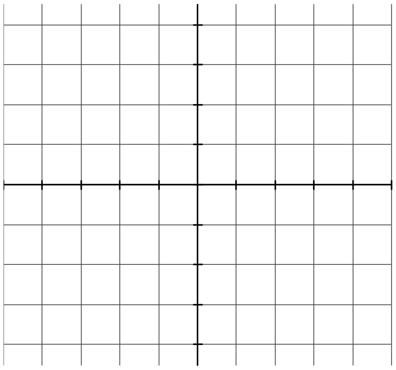
13. $y = \sec x$



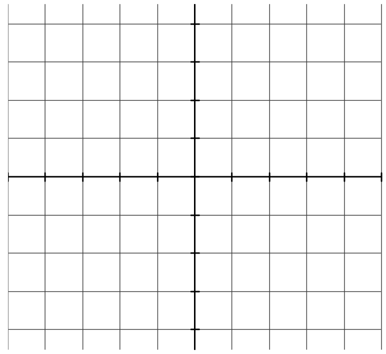
14. $y = \csc x$



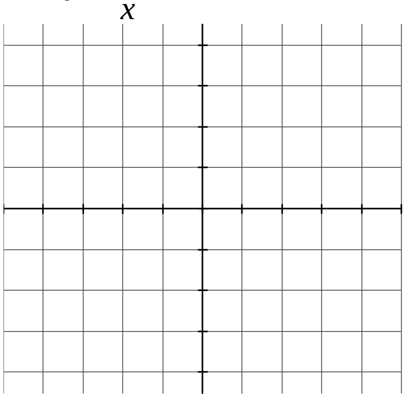
15. $y = e^x$



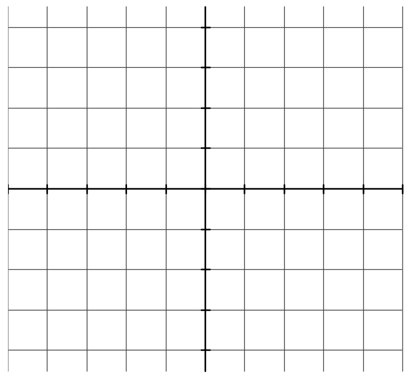
16. $y = \ln x$



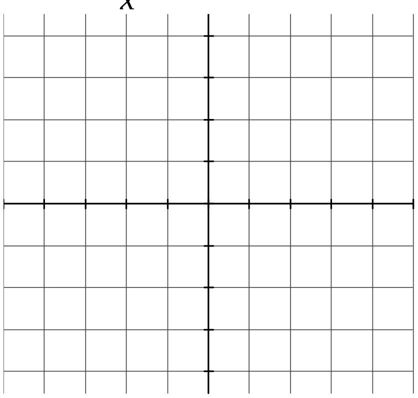
17. $y = \dfrac{1}{x}$



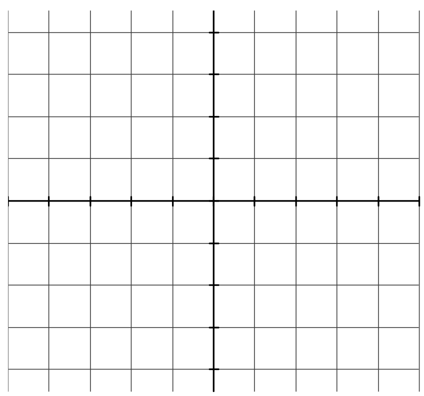
18. $y = |x|$



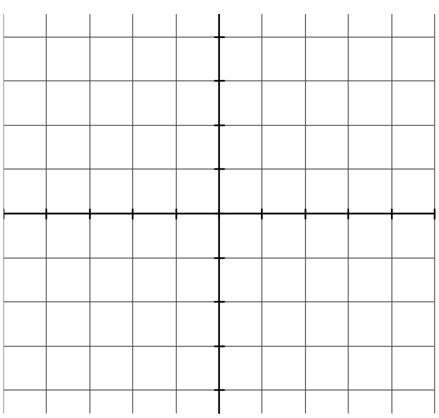
19. $y = \dfrac{1}{x^2}$



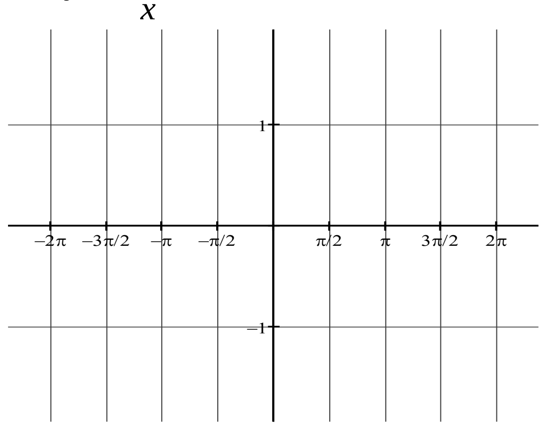
20. $y = 2^x$



21. $y = \sqrt{4 - x^2}$



22. $y = \dfrac{\sin x}{x}$



Topic D: Function Transformations

If $f(x) = x^2 - 1$, describe in words what the following would do to the graph of $f(x)$:

1.) $f(x) - 4$

2.) $f(x-4)$

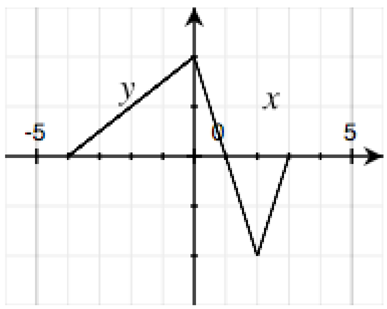
3.) $-f(x+2)$

4.) $5f(x) + 3$

5.) $f(2x)$

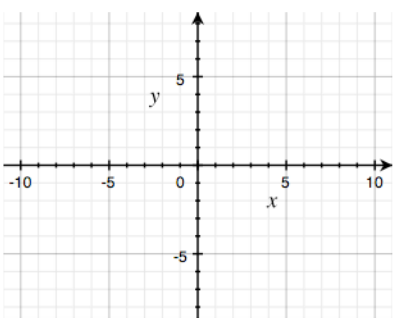
6.) $|f(x)|$

Here is a graph of $y = f(x)$:

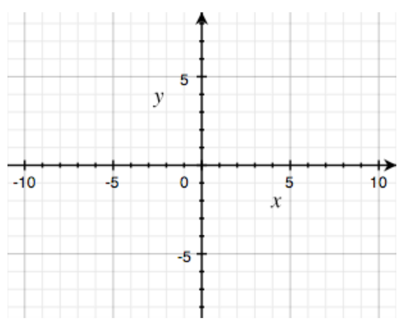


Sketch the following graphs(hint: make tables of values)

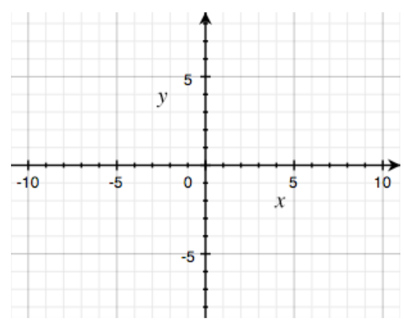
7.) $y = 2f(x)$



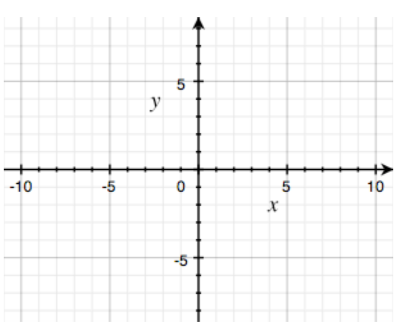
8.) $y = -f(x)$



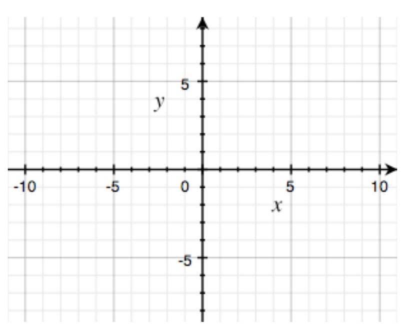
9.) $y = f(x-1)$



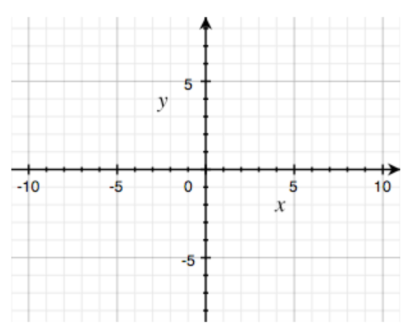
10.) $y = f(x) + 2$



11.) $y = |f(x)|$



12.) $y = f(|x|)$



Topic E: Special Factorization

Factor completely.

1.) x^3+8

2.) x^3-8

3.) $27x^3-125y^3$

4.) x^4+11x^2-80

5.) $ac+cd-ab-bd$

6.) $2x^2+50y^2-20xy$

7.) $x^2+12x+36-9y^2$

8.) $x^3-xy^2+x^2y-y^3$

9.) $(x-3)^2(2x+1)^3+(x-3)^3(2x+1)^2$

Topic F: Linear Functions

1.) Find the equation of the line in point-slope form $y-y_1=m(x-x_1)$, with the given slope, passing through the given point.

a.) $m=-7,\ (-3,-7)$

b.) $m=-\frac{1}{2},\ (2,-8)$

2.) Find the equation of the line in point-slope form, passing through the given points.

a.) $(-3,6),\ (-1,2)$

b.) $(-7,1),\ (3,-4)$

3.) Find the equations of the lines through the given point that are a.) parallel and b.) perpendicular to the given line.

a.) $(5,-3),\ x+y=4$

b.) $(-6,2),\ 5x+2y=7$

Topic G: Solving Quadratic and Polynomial Equations

Solve each equation for *x* over the real number system.

1.) $x^2+7x-18=0$

2.) $x^2+x+\frac{1}{4}=0$

3.) $2x^2-72=0$

4.) $12x^2-5x=2$

5.) $20x^2-56x+15=0$

6.) $81x^2+72x+16=0$

7.) $x+\frac{1}{x}=\frac{17}{4}$

8.) $x^3-5x^2+5x-25=0$

9.) $2x^4-15x^3+18x^2=0$

Topic H: Asymptotes

For each function, find the equations of both the vertical asymptote(s) and horizontal asymptote (if it exists) and the location of any holes.

1.) $y=\dfrac{x-1}{x+5}$

2.) $y=\dfrac{8}{x^2}$

3.) $y=\dfrac{2x+16}{x+8}$

4.) $y=\dfrac{2x^2+6x}{x^2+5x+6}$

5.) $y=\dfrac{x}{x^2-25}$

6.) $y=\dfrac{x^2-5}{2x^2-12}$

7.) $y=\dfrac{x^3}{x^2+4}$

8.) $y=\dfrac{x^3+4x}{x^3-2x^2+4x-8}$

9.) $y=\dfrac{10x+20}{x^3-2x^2-4x+8}$

Topic I: Negative and Fractional Exponents

Simplify and write with positive exponents.

1.) $-12^2 x^{-5}$

2.) $\left(-12x^5\right)^{-2}$

3.) $\left(4x^{-1}\right)^{-1}$

4.) $\left(\frac{-4}{x^4}\right)^{-3}$

5.) $\left(\frac{5x^3}{y^2}\right)^{-3}$

6.) $\left(x^3-1\right)^{-2}$

7.) $\left(121x^8\right)^{1/2}$

8.) $\left(8x^2\right)^{-4/3}$

9.) $\left(-32x^{-5}\right)^{-3/5}$

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Topic J: Exponential Functions and Logarithms

Simplify the following without a calculator:

1.) $\log_2 \frac{1}{4}$

2.) $\log_8 4$

3.) $\ln \frac{1}{\sqrt[3]{e^2}}$

4.) $5^{\log_5 40}$

5.) $e^{\ln 12}$

6.) $\log_{12} 2 + \log_{12} 9 + \log_{12} 8$

7.) $\log_2 \frac{2}{3} + \log_2 \frac{3}{32}$

8.) $\log_{\frac{1}{3}} \frac{4}{3} - \log_{\frac{1}{3}} 12$

9.) $\log_3 \left(\sqrt{3}\right)^5$

Solve the following:

10.) $\log_5 (3x-8) = 2$

11.) $\log_9 \left(x^2 - x + 3\right) = \frac{1}{2}$

12.) $\log(x-3) + \log 5 = 2$

13.) $\log_2 (x-1) + \log_2 (x+3) = 5$

14.) $\log_5 (x+3) - \log_5 x = 2$

15.) $\ln x^3 - \ln x^2 = \frac{1}{2}$

16.) $3^{x-2} = 18$

17.) $e^{3x+1} = 10$

18.) $8^x = 5^{2x-1}$

Topic K: Basic Right Angle Trigonometry

Solve the following:

If point P is on the terminal side of θ, find all 6 trigonometric functions of θ. (Answers need not be rationalized.)

1.) $P(-2,4)$

2.) $P\left(\sqrt{5},-2\right)$

3.) If $\cos\theta=-\frac{5}{13}$, in quadrant II,
find $\sin\theta$ and $\tan\theta$.

4.) If $\cot\theta=\frac{2\sqrt{10}}{3}$, in quadrant III,
find $\sin\theta$ and $\cos\theta$.

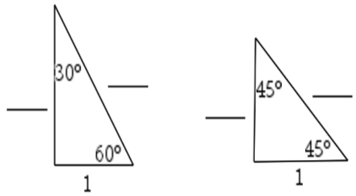
5.) State the quadrant in which each of the following is true.

a.) $\sin\theta>0$ and $\cos\theta<0$

b.) $\csc\theta<0$ and $\cot\theta>0$

c.) $\tan\theta>0$ and $\sec\theta<0$

6) Fill in the four blanks.



Topic L: Trigonometric Identities

Verify the following identities:

1.) $(1+\sin x)(1-\sin x)=\cos^2 x$

2.) $\sec^2 x+3=\tan^2 x+4$

3.) $\dfrac{1-\sec x}{1-\cos x}=-\sec x$

4.) $\dfrac{1}{1+\tan x}+\dfrac{1}{1+\cot x}=1$

Topic M: Solving Trigonometric Equations

Solve each equation on the interval $[0,2\pi)$. Do not use a calculator.

1.) $\sin^2 x=\sin x$

2.) $3\tan^3 x=\tan x$

3.) $\sin^2 x=3\cos^2 x$

4.) $\cos x+\sin x\tan x=2$

5.) $\sin x=\cos x$

6.) $2\cos^2 x+\sin x-1=0$

Topic N: Using the Graphing Calculator (Continued)

Use your graphing calculator to find the zeros of each of the following functions. Make sure each equation is set equal to zero first.

1.) $3x^3 - x - 5 = 0$

2.) $2x^2 - 1 = 2^x$

3.) $2\ln(x+1) = 5\cos x$ on $[0, 2\pi)$

Use your graphing calculator to find the solution (intersection) of the given system of equations.

4.) $\begin{cases} f(x) = x^4 - 6.5x^2 + 6x + 2 \\ g(x) = 1 + x + e^{x^2 - 2x} \end{cases}$

Use your graphing calculator to find both a relative maximum and a relative minimum point of the given function.

5.) $h(x) = 2x^5 - 3x^4 + x - 4$